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1. **Question: How long after the Lane Departure Warning System does the Lane Keeping System activate?**

- It needs to be event based, immediate, at your discretion, if there needs to be a time delay or warning delay. Based on the event of it leaving the lane.
- immediate, at your discretion if there needs to be a time/warning delay - event should be that it is leaving the lane (determine what is the definition of leaving the lane)
- In the event of a lane departure: immediately.
- Takes an event based amount of time to activate, immediate if it needs to be, can have a time delay
    - Based on leaving the lane, not prediction
- LKS activates when car actually leaves the lane

2. **What defines a momentary or partial lane departure? :**

- Consider vehicle boundary and lane boundary and LMS system should be aware of its vehicle boundaries and if that is crossing the lane boundary than its a full departures. Partial is touching or approaching it. Assume the boundaries are known at a give point.
- Vehicle boundary/lane boundary - LMS system should be aware of these, if this is crossing either of the boundaries then it is a full departure, and partial would be touching or approaching it.
- If a vehicle boundary crosses a lane boundary the system recognizes this as a departure. If a boundary of the vehicle is touching or approaching a lane boundary the system recognizes this as a partial departure.
- Should be aware of the vehicle boundary
    - Can have partial or full departure of the lane
    - Partial = touching lane or approaching lane
- Crossing lane departure activates it

3. **At what point can be declared the confidence is low enough to hand over the control, is there a variable that has to be above a certain value? Is there an amount of force the driver needs to apply to let the LMS know the driver wants to take over?**

- The driver taking control is separate. LMS by itself should proceed to bring the car back into the lane and it should keep doing that it should not voluntarily give up control unless there is other interference. Apply an amount of torque on the steering wheel we can determine our own teams threshold for the driver to take back over, varies depending on the speed of the vehicle.

- LMS should proceed with bringing car back to the lane and should not momentarily give up control to the user, unless the driver interferes on their own
- Is there a certain threshold?
    - Application of torque on steering wheel can determine control hand off
    - Speed is also taken into consideration
- With no interference from the driver through applying torque on the steering wheel or manual deactivation the LMS should always remain activated.
- LMS should not voluntarily give up control
    - Unless the driver interferes to take control back
        - A certain amount of torque on the steering wheel is the threshold needed to overtake control (speed dependent)

4. **How should the Lane Management System respond in areas where the lanes are not clear or ambiguous?**

- As a prereq markings need to be visible, or there needs to be enough information for it to be enabled. Also there is map data for roads which can be used in place of the cameras. If there is not enough information LMS should be inactive. The driver should be able to know that this is disengaged. The driver should always know if it is active or inactive.
- Needs sufficient data for LMS to be active, all markings need to be visible as a prerequisite. If any of this seems to not be available then LMS must be inactivated and the driver should be aware of it being disengaged.
- How should the driver know whether its inactive or active?:
    - Cluster light or any other means deem fit
    - Anywhere in the driver console
- If lane markings cannot be identified, the LMS will be inactive
- Needs to be sufficient data for LMS to be active
    - Has map data, of roads and lanes
    - If its not sufficient, LMS should be inactive and the driver should know its disengaged
        - Cluster light, driver console, any other means to show driver its inactive
- LMS inactive if road lanes aren't detected

5. **How should the driver know if the LMS is active or inactive?**
    - a light, any other means we see fit, HUD, driver console, dash board, etc.
    - Needs sufficient data for LMS to be active, all markings need to be visible as a prerequisite. If any of this seems to not be available then LMS must be inactivated and the driver should be aware of it being disengaged.
    - How should the driver know whether its inactive or active?:
        - Cluster light or any other means deem fit

    - Anywhere in the driver console
- Symbols displayed on the driver's console/heads-up display, or cluster lights are possible options. It's up to your discretion which option that you choose.
- When lane keeping is active (taking control of wheel), sound alerts, blinking lights, haptic feedback to show its active
    - Don't want to over notify the driver (minimal to a point)

6. **What kind of systems or features does the car have to indicate warnings?**
    - There can be driver sound alerts, blinking lights, cluster alerts, haptic feedback on steering wheel or seat. If we have others they can incorporate that. We do not want to over notify the driver, we hae to consider ways where it is minimal to a point.
    - Driver sound alerts, or blinking light/cluster alerts
    - Haptic feedbacks on steering wheel or seats
    - Do not want to overnotify the driver, have to consider ways where it is minimal to a point

7. **What is the main difference between LKS and LCS?**
    - LCS is the system that is meant to keep the vehicle in the center line of the lane and that has active steering control and chnge the position of the vehicle but it should have its own limitations on where it can move cause its only working within a lane boundary. LKS is the one that activates to move the car back into a lane
    - LCS third system that keeps car in center line, active steering control needed, it can still change position of the vehicle but needs limitations on how it can move the vehicle
    - Helps to notify the other systems of out of bounds with the vehicle and lanes
    - LCS should keep the vehicle centered in the lane. It can change the position of the vehicle but it should have limitations on how much it can do so. The Lane Keeping System is activated specifically to move the car back into the lane.
    - Difference between lane keeping system and lane centering system
        - Centering system = keeping vehicle in center of lane
        - Keeping = keeping vehicle in the lane at all
        - Keeping is stronger than centering

8. **Do brief absences in lane markings produce the same behavior?**
    - Brief absences can be covered if there is map data available. So lane inference and vehicle path could be determined. At different speeds a brief absence can be different. If it blocks the lane inference and path then LMS should go inactive.
    - Brief absences can potentially look different at different speeds

- If the absence of lane markings is large enough that the path cannot be determined, LMS is deactivated.
- Brief absence of lane markings is covered if map data is available
    - Brief absence is speed dependent
- If map data is available, lane and vehicle position can be determined, speed can change determination

9. **What would you describe being the max turn radius the system should be able to turn the car?**
    - There is no set limit, it is determinded by the speed of the car and the curve of the road, it should be able to adapt to standard road conditions. Even on highways and that fast speed.
    - No set limit, determined on the speed and curvature of roads, should be able to adapt to normal driving conditions and standard roads.
    - There is no set limit. The amount of turn radius is determined based on vehicle speed and curve of road.
    - No set limit, speed based and dependent and curve of the road
    - Should be accountable for highways

10. **Is there any time we want to system to be active or inactive for when the vehicle does not want to be active, for laws or regulations?**
    - Initial design we do not want to incorporate this. Laws he can think of is vehicles operating in different countries with reverse lanes or other sides of the road.
    - No incorporation needed as of now
    - To incorporate this they'd have to capture specific city law data.
    - Other laws that can be considered is vehicles in other countries with reverse lanes.
    - It does need to be a focus in the design.
    - Different countries, reverse lanes needing system to be treated differently

11. **How will an intentional departure from a lane be defined?**
    - Only with a turn signal. And the vehicle leaving the lane boundary should match the direction of the turn signal.
    - Turn signal and vehicle movement to match turn signal
    - If the turn signal matches the direction in which the vehicle leaves the lane boundary, the system considers this an intentional departure.
    - If the turn signal is switched one direction but the driver steers the vehicle outside the boundary in the opposite direction, what should happen?
        - Torque applied from the driver has primary control over the vehicle path. A lane departure warning will occur.

- Only with the turn signal, must match the direction of the turn signal as you're moving
    - Threshold is primary even above the turn signal, can still show departure warnings
- The threshold is still primary → still warnings, no lane shift.

12. If the car is moving the wrong direction of the turn indicator but its purposeful torque which do we follow?

- The drivers torque threshold. The driver would still get the departure warnings in this case.

13. How would LDWS react to other vehicles entering into the drivers lane.

- Out of scope.
- Out of scope, scope for LMS is to keep vehicle in its lane, if other vehicles are coming in, that may cause the driver to swerve.
- Out of scope of the system. LMS does not handle interactions with other vehicles.
- Other vehicles are out of scope for this project

14. How should the LMS behave in different weather conditions?

- The lane path needs to be able to be inferred via camera and map if it's not there then it needs to go inactive.
- Basic requirement would be that the lane path would be inferred via the cameras or lane system
- Basic requirement of identifying lane markings still apply

15. How much control would you like for the driver to have in terms of utilizing the Lane Management System?

- The can turn on and off during driving as LMS as a whole and turn off LKS separately. They need to be able to see the status of this in some lane display. Only LKS has specific shut off control.
- Can turn on off during driving, can turn it on or off as a whole or specifically LKS as well, can keep the warnings but no override.
- Need to be able to see the status in some vehicle display
- Can some be disabled or enabled?
    - Only LKS has specific control - others basically built into LMS
- They can activate and deactivate the LMS while driving.
- Only LKS can be individually deactivated

16. **How would the LMS respond to older lines on the road, old lines that are not the current road.**
    - Even during construction we expect LMS to be active there should be a lane or road boundaries that are present durng that time and see the new lines, if there is confusion or multiple lines it should go inactive.
    - Expect LMS to be active, there should be a lane or road boundary that is present at that time, where LMS can still identify lane markings
    - If there is confusion it should go inactive
    - Even during construction LMS should work
        - Inactive if there's confusion

17. **What conditions do the sensors and camera for LMS to function properly? Is there a healthy check that needs to be passed for LMS to turn on?**
    - You can assume that as part of the hardware components, there should be an overall all cameras active check that is there. If any of them are missing then that should signal the LMS to be inactive. All cameras must be active. In the event that a camera becomes in active while driving LMS will need to become inactive.
    - Can be assumed as hardware components, if any missing then the system should be inactive
    - If there is a camera not function properly should LMS still activate?
        - All cameras need to be available
    - If the camera becomes unavailable halfway through driving
        - Immediately become unavailable (standard rule)
    - Has an all cameras active check
        - Only if all cameras are active and working does LMS activate, otherwise deactivate
            - Can think of a backup option
    - If camera becomes inactive while driving, immediately LMS go inactive with warnings

18. **Should the driver be able to override the Lane Keeping System with enough force or should there be another method to override?**
    - Driver could manually turn off LMS at that time, even during engagement.
    - Driver can turn off LMS even during engagement.
    - Can turn it off even during driving
    - Can turn off the keeping but keep the warnings
        - Can ONLY turn off the LKS specifically
    - During engagement, driver can still deactivate LMS

19. How does the LMS distinguish if the en do the lane is gravel, not painted lines

- Can be part of the camera subsystems, LMS needs an input of the lane boundary. If gravel or wall can be treated as the lane boundary it can work
- Can be part of camera and hardware subsystems
- LMS needs an input of lane boundary
- Can treat gravel or walls as the lane boundary

20. How should the vehicle respond to the driver taking their hands off of the steering wheel?

- Up to our discretion if we need that. The intention is not autonomous driving. Would be in scope to keep driver attention.
- The intention of LMS is not autonomous driving but to keep vehicle in the lane, so still sticking to the main purpose of lane management
- Should be in scope: Keeping driver's attention.
- Up to our discretion, in scope

21. Will the Lane Management System automatically be turned on once the driver begins to drive, or will this have to be manually turned on?

- Should resume the last state from the previous ride. If off it stays off if on it turns on.
- Resume action from previous rides. If it is off it should remain off.
- It should resume the activation state from the last time the vehicle was driven.
- If you keep it off, it wont turn on on next engine start and vise versa
- On drive startup, it should follow the last LMS settings from last vehicle use

22. How should the LMS communicate with the GPS?

- Camera system just provides live camera data. GPS will give you vechile coordinates and vehicle speed and heading directionn at every one second, the coordinates can be used in conjunction with the map data that has lane boundaries at coordinates. 4
- Camera system keeps track of lanes ahead
- GPS gives vehicle coordinates and vehicle speed & heading direction
- Coordinates should be in conjunction with map data
- If GPS is inactive or unusable how should LMS respond?
    - It doesn't need GPS, only required when Map system is needed as well.
- The GPS system will provide vehicle coordinates, vehicle speed, and direction every 1 seconds.

- GPS gives vehicle coordinates, vehicle speed, heading, and direction
    - Every one second
- Map data has lane boundary at coordinates
- If GPS becomes inactive, LMS does not turn off
    - LMS can go off of cameras alone

23. If GPS becomes inactivate what happens?

- It does not need the GPS it can work just on cameras as long as the cameras provide enough data.
- The LMS does not need both to operate. It can operate using just the cameras.
- Doesnt need gps and cameras, can run on just cameras

24. Whenever the driver turns the turn signal on and once it turns off how long should the LMS wait to turn back on?

- Immediately after turn signal is off
- Instantaneous, only inactive during turn signal
- Instantaneous once turn signal turns off

25. Between what speed thresholds should the Lane Management System be active?

- At over 35 miles per hour and under 85 miles per hour, exclusive
- 35mph - 85mph (exclusive)
- Over 35 mph and under 85 mph (exclusive).
- Active over 35 mph
- Under 85 mph
- Exclusive
- Active between 35-85 MPH, exclusive

26. Is there a certain terrain or road that LMS should always be inactive?

- No it is always trying to detect
- Should always be trying to detect regardless of road type

27. How should the torque quantity change based on different weather conditions?

- Up to our discretion to avoid slip hazard
- Up to our discretion
- Should be useful to avoid slip hazards
- Up to our discretion
- To avoid slip hazards

- Torque applied during different conditions should avoid slip, up to our discretion

28. How will the Lane Management System behave at different speed thresholds?

- Should be adjusting the amount of torque that LMS applies to the vechile, at higher speeds you need to adjust how much influence LMS has on the vehicle to keep it in the lane. The camera system will need to capture further ahead in the lane to be able to inference faster in the same distance.
- Should be adjusting amount of torque that LMS applies to the vehicle
- Higher speeds = lower effort (adjust how much influence LMS has on the vehicle)
    - Camera system: capture further ahead in the lane to be able to inference faster
- It will adjust how much torque is applied to vehicle steering. Less torque is needed at higher speeds and more is needed at lower speeds.
- The camera system will capture images further ahead in its path.
- Adjust amount of torque LMS applies, higher speeds = lower torque
- Camera system captures further ahead at higher speed
- Torque should be adjusted at different speeds

29. For camera capturing system, what speed does it capture images?

- Out of scope, just need to determine how many meters ahead it needs to capture.
- Out of scope
- At faster speeds camera needs to capture lane from further ahead

30. How should the LMS behave when the vehicle is exiting or entering a highway and needs to lane merge?

- If its on its keeping the car in lanes so it would behave normally. Map data will contain as a merging lane and it could be excluded but this is a out of scope.
- Searching and keeping it in lane, behaves normally
- Can exclude merge lanes
    - Should have turn signal on anyway

31. How will the relative positions of the vehicle within a lane be determined?

- LMS knows based on the vehicle boundary and with its a combination of the GPS and the camera subsystem to see that. The relative position would be proritezed by the centering system, if the data from camera and GPS that centering system would use that to figure out that it is off center.

- LMS knows vehicle boundary
- It is a combination between the camera and the gps
- Relative position prioritized by the centering system
- Is there a lateral offset that should trigger the lane system?
    - At your discretion what offset is needed, based on the vehicles can potentially have a smaller offset.
- The LMS determines whether the vehicle is centered in the lane by using the vehicle boundaries and the lane boundaries.
- Should be determined by vehicle boundary, gps, and camera
- The offset that is needed for LCS to activate is at our discretion
- Lateral offset determined by our group – relative to vehicles size

32. Is there a center horizontal offset that should trigger LCS?

- At our discretion what offset is needed. Based on the vechiles that can also have potentially have a smaller offset than others.
- At your discretion. One good idea is to have the offset be based on the vehicle's size.

33. Do different sensors ever disagree on what the vehicle is doing, how should the LMS behave?

- Treat each system as its own and its has its own prereqs to activate it so they shouldn't be competing. Difference between camera and GPS, cameras are the priority and map is just secondary to help so make camera as the priority .
- Can treat each system as its own duties, LKS has its own prerequisites that trigger it and so on.
- What if maps thinks the construction zone has a different direction than reality?
    - Cameras are priority, map is redundant second priority
- Treat each system as its own duties with different prerequisites
- Camera and GPS disagreement
    - Cameras take priority, maps are secondary as a redundant
    - Map can't be relied on
- Cameras take priority over the map system

34. How will the Lane Keeping System alter the vehicle's path?

- Sending signals to the steering wheel and the speed, breaking, acceleration.
- Based on steering wheel and speed, based on if it is breaking acceleration.

- Steering wheel and speed are affected by the system

35. How sould the LMS react to different vehicles?

- LMS does account for what vehicle it is in, but this is out of scope, different speed different torque, different vehicle boundary.
- Out of scope but does account for the difference in vehicle
- Vehicle boundary, speed, and torque (class 1,2,3 and size of vehicle)

36. Does the Lane Centering system make adjustments to the vehicle even if the lane hasn't been left?

- Would only use steering and its events trigger is the offset of the center line, very minimal amount of steering control, should have guard rails.
- Would only use steering, its events to trigger this would be offsets off the center line
- Consider that it is a very minimal amount of steering
- When should the LMS understand that it needs to apply a torque?
    - Whatever percentage of offset is from the center line is when it should be applied
- Lane centering only use steering, lateral offset

37. How should it be determined what speed to recenter? LCS

- That would be based on event based, off of the offset of the center line, if you breach the threshold it applies.
- Event based - off of the offset of the center line

38. To what degrees will the Lane Keeping System be able to alter the vehicle's path?

- Yes it should have guard rails where it can not exceed a certain amount of vehicle safe so it doesn't 90 degree turn the car. Guard rails can be determined by our research. Can be researched the speed limits as well.
- Can be determined by research. Guard rales should be in place

39. Should there be a threshold where camera visibility isnt the best and the system should start relying on GPS or maps?

- No threshold it either is showing the lines or not. Out of scope. Camera would rely on a confidence interval but we can treat it as yes or not
- No threshold, it either shows the lines or not
- Out of Scope
- Confident interval needed for camera detection

40. Is it possible for both the Lane Keeping System and Lane Centering System to be active at the same time?

- Should not be at the same time, keeping the system would take priority in any edge case. They work together keeping system brings it back then centering takes over.
- Should not be at the same time, LKS should take priority.
- No they should not be active at the same time. When an edge case is reached, LKS should be activated to steer the vehicle back into lane. Once back in lane, the LCS resumes activation to center the vehicle.
- Can LCS and LKS be active at the same time
    - No, LKS takes priority
    - Keeping brings it back into the lane, centering is within lane
    - LKS takes priority in edge cases over LCS

41. What should happen after the driver turns LMS off, how long does it take LMS to actually turn off? Does it keep gathering data?

- Gather data can still be active, with respect to the subsystems that disable completely.
- Gathering data can still be active, with respect to other systems, it should be disengaged.

42. What should be in the logs?

- all events taken by LMS need to be logged. The log comes with the vehicle state, speed headings, driving conditions, what triggered the event, what the event was, all warnings, all overrides.
- Logging, can determine what specific logs you want but all events taken by LMS need to be logged, and log comes with vehicle state
- Logs should have all events taken, vehicle state (speed, heading), what triggered LMS event, what action was taken, was it overridden
- Logs everything headings as well as decision making

43. What data if any is being sent off the vehicle?

- All the logs are collected, sent off can mean it collected into the cloud system that the vehicle may have. This would not be available to the driver. Any camera or system health data should be logged.
- Data logged in vehicles cloud, not available to user

- All events taken by LMS should be logged, along with the vehicle's state (telemetry, warnings)
- Log data not available to the driver
- Some data (corrections and warnings) are sent off

44. General warning advice

- do not over alert the driver we need to design something that is there and more silent, if it gets annoying the system will be turned off and it wouldnt get used.
- Minimal warnings, other systems on the vehicle that have their own warnings so be cautious about how many warnings the system gives off, too many warnings makes the driver turn the system off
- Minimal warnings, don't over alert with just LMS warnings
- Don't be annoying
- Don't over notify